Applied Software Project Report

By

<Full Name of the Student>

A Master's Project Report submitted to Scaler Neovarsity - Woolf in partial fulfillment of the requirements for the degree of Master of Science in Computer Science

<Month of Submission like June, 2024>

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Thesis Supervisor : Naman Bhalla

Date of Submission : DD/MM/YYYY <Date of Submission>

© The project report of <Full Name of the Student> is approved, and it is acceptable in quality and form for publication electronically

Certification

I confirm that I have overseen / reviewed this applied project and, in my judgment, it adheres to the appropriate standards of academic presentation. I believe it satisfactorily meets the criteria, in terms of both quality and breadth, to serve as an applied project report for the attainment of Master of Science in Computer Science degree. This applied project report has been submitted to Woolf and is deemed sufficient to fulfill the prerequisites for the Master of Science in Computer Science degree.

Naman Bhalla

……

Project Guide / Supervisor

DECLARATION

I confirm that this project report, submitted to fulfill the requirements for the Master of Science in Computer Science degree, completed by me from < Project Module start date > to < Module end date >, is the result of my own individual endeavor. The Project has been made on my own under the guidance of my supervisor with proper acknowledgement and without plagiarism. Any contributions from external sources or individuals, including the use of AI tools, are appropriately acknowledged through citation. By making this declaration, I acknowledge that any violation of this statement constitutes academic misconduct. I understand that such misconduct may lead to expulsion from the program and/or disqualification from receiving the degree.

<Full Name of the Candidate>

<Signature of the Candidate> **Date: XX Month 20XX**

ACKNOWLEDGMENT

<Insert a Paragraph within this Page to express gratitude to your family, Scaler instructors and everyone who helped, inspired or motivated you to complete the program and earn the Master's degree>

Table of Contents

List of Tables **6**
List of Figures **7**
Applied Software Project **8**
Abstract 8
Project Description 8
Requirement Gathering 9
Payments Integration 9
Security 9
Deployment Flow 10
Technologies Used 10
Conclusion 10
References **11**

List of Tables

(To be written sequentially as they appear in the text)

Table No.	Title	Page No.
1		
2		

List of Figures

(List of Images, Graphs, Charts sequentially as they appear in the text)

Figure No.	Title	Page No.
1		
2		

Applied Software Project

Abstract

The abstract for an applied software project should succinctly summarize its title, purpose, methods, results, and conclusions.

Since it is a technical project, focus on its real-life applications in the society or how such software tools can be applied to modify existing processes for better outcomes in different industries.

Aim for conciseness, usually limiting it to under 300 words, to effectively convey the project's essence.

Project Description

Describe the project, Outline the objectives and explain its relevance.
Use Pictures, flow diagrams to illustrate the project well

Use the below format as reference when including a picture

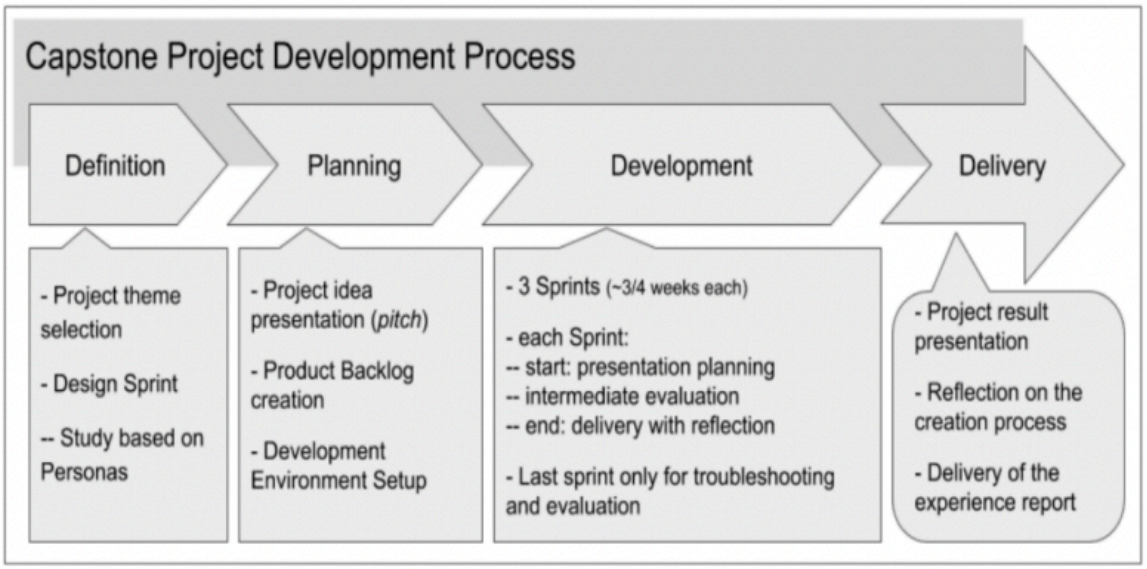

Figure 1.1: Project Development Process (Figure captions go below figures.)

Requirement Gathering

- Describe the requirements - Functional, Non-Functional Requirements
- Describe the Users and Use Cases
    - Include detailed Use Case Diagram / Images to illustrate
- List and detail out the Feature set -
    - Use Tables to list out the feature set

Use the below format as reference when including a table

Table 1.1: < Table caption > (Table captions go above tables.)

Payments Integration

Explain how Payment Gateways work, describe the payments flow, discuss the security and standards followed (PCI DSS)

Begin by selecting a suitable payment gateway like Razorpay, Stripe
Lookout for integration related code in the docs like for Razorpay -
https://razorpay.com/docs/payments/server-integration/nodejs/payment-gateway/build-integration/

React specific integration -
https://razorpay.com/docs/payments/server-integration/nodejs/faqs/#3-can-i-integrate-razorpay-checkout-with-reactjs

- Role of webhooks, HTTP Tunneling (used for webhook when backend running locally)

Security

a. Discuss bcrypt. discuss salt and encryption . using bcrypt for password hashing
b. Different node packages for added security measures like express-rate-limiter (api rate limiting), express-mongo-sanitize (for no-sql injection) , helmet (hiding server details

in response), etc

Deployment Flow

a. Maintain code in github
b. Can use aws or for simple deployment process, can use Render
c. For frontend deployment many use Netlify. If you do so, then can use Render for backend apis and then connect to it through URL in Netlify
d. Remember to handle CORS issue

Technologies Used

MERN Stack -

Mongo, React, NodeJS, Express (MERN Stack) - Detail each of them and other technologies that have
been used if possible with diagrams and also elaborate how they can be used in real life Applications by giving examples….
(Include proper references if you are quoting articles from the internet)

Tip - Use the internet to improve and enrich your project but DO NOT PLAGIARIZE - Include proper references if you are quoting articles from the internet

Conclusion

The Conclusion should include some key points as elaborated below -

- Key Takeaways: Highlight the important concepts and technologies learned from doing the Project
- Practical Applications : Significance of technologies with their real-world applications
- Limitations : Limitations of the technologies, cost implications and suggestions for improvement

References

Include the websites or works or the list of works referred to in a text or consulted by you for writing this report

1. Name of the Website, Date and time of referring to the Website, Name of the Author, Title/Topic
2. Author Name, Title / Topic, Research Paper Name / Book Name, Year of Publication

Format Guidelines

1) Detailed and Elaborate report of 40 pages at least is expected
2) Margins - Every page of your document must meet the margin requirements of 1.25 inches on the left and right, and 1 inch on the top and bottom.
3) Font:
    a) Style: Times New Roman,
    b) Font Size:14 (For Headings), 12 (For body text) in black colored text.
    c) All text must be the same justification, like left justified or fully justified.
4) Line Spacing:
    a) Body of the text: 1.5
    b) List of Tables/graphs/charts/bibliography: single line.
5) Alignment:
    a) Title page: Centre
    b) Chapter Heading: Centre
    c) Subheading: Left
    d) Body of the text: Justify
6) Titles: All titles and subtitles should be in bold. All tables/graphs/charts/figures should have appropriate titles.
7) Numbering of the tables, charts, graphs should be in the following fashion: Second table/graph/chart in the second chapter should be numbered as Table/graph/chart no. 2.02; where the first digit stands for chapter no. and digits after (.) stands for number of table/graph/charts in that chapter. Same numbering should be followed for all other chapters.